

Agentic-J chose BoneJ and reproduced Table 2 of the paper: 3 of 4 metrics within 0.9%, Tb.Sp +5 to +10% from an unreported setting

Task: quantify sarcomere orientation in hiPSC-derived cardiomyocytes, marked by α-actinin

Agentic-J chose TrackMate: 29 tracks, all on published cells; mean speed 2.00 vs 2.02 µm/min published, r = 0.99

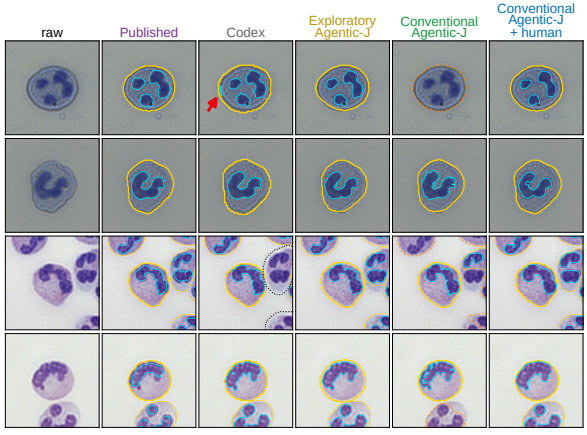
d

Analysis	Time	Model output	Scientific verdict
CD177 neutrophil morphology			
Published	weeks	–	no meaningful difference
Codex CLI	13 min	0/30 features significant	no biologically meaningful conclusion due to faulty segmentation
Exploratory Agentic-J	35 min	5/25 significant, 3/3 runs	no biologically meaningful conclusion due to faulty segmentation
Conventional Agentic-J	67 min	0/32 significant, 3/3 runs	no biologically meaningful conclusion due to faulty segmentation
Conventional Agentic-J + human	151 min	0/7 features significant	no meaningful difference detected, needs validation

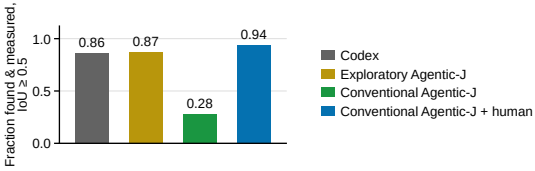
Time T1 / T2	Task 1 TUNEL+ and TUNEL− nuclei	Task 2 proliferation in TUNEL+ hosts
Leishmania infection: TUNEL and proliferation		
not recorded	TUNEL+: nuclei 3, 6 TUNEL−: nuclei 1, 2, 4, 5	higher proliferation in TUNEL+ hosts 8 TUNEL+ vs 7 TUNEL− parasites
18 min, both tasks	TUNEL+: nuclei 3, 6 TUNEL−: nuclei 1, 2, 4, 5	higher proliferation in TUNEL+ hosts Δ median 1.68; 10 of 15 parasites
22 / 69 min	TUNEL+: nucleus 3, split in two TUNEL−: nuclei 1, 2, 4, 5, 6	not testable: no parasite found 22 false detections, no host
55 / 85 min	TUNEL+: nucleus 3 TUNEL−: nuclei 1, 2, 4, 5, 6	not testable: no TUNEL+ host found 3 of 15 parasites, 2 false
35 / 45 min	TUNEL+: nuclei 3, 6 TUNEL−: nuclei 1, 2, 4, 5	higher proliferation in TUNEL+ hosts Δ median 2.36 (published data 2.36); 15/15

Manual (analyst) · Human feedback (user) · Automatic (Agentic-J) · Automatic (other agent)

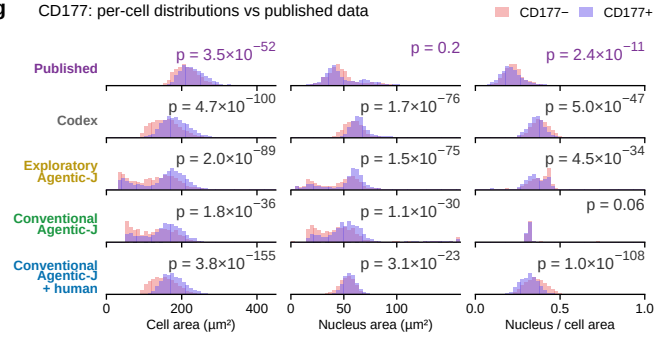
e CD177: segmentations of four example cells



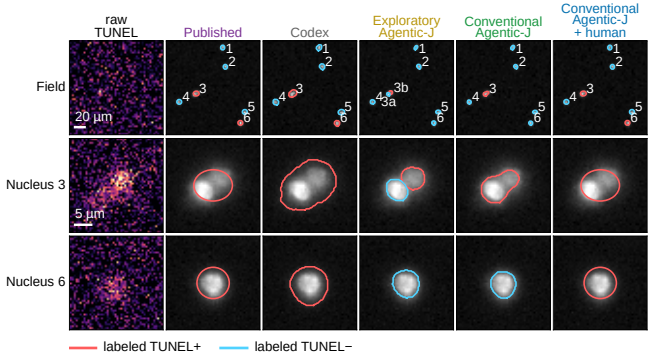
f CD177: published cells found and measured, n = 4389



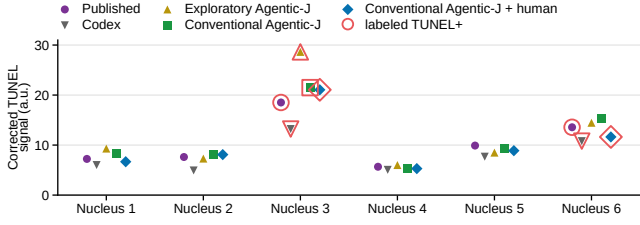
g CD177: per-cell distributions vs published data



h Leishmania task 1: six nuclei, called by each pipeline



i Leishmania task 1: corrected TUNEL signal per nucleus



j Leishmania task 2: proliferation index by host TUNEL status

